IQP Generative Models for Strong Gravitational Lensing Images

Anonymous Author(s)
Affiliation
Address
email

Abstract

Instantaneous-quantum-polynomial (IQP) circuits are believed hard to sample classically [5, 6], yet the expectation values of a maximum-mean-discrepancy (MMD) loss stay computable, so an IQP Born machine can be trained classically and deployed on quantum hardware [24]. A recent correlation–complexity map is proposed to indicate which data suit such models, and DeepLense strong-lensing images sit at its most IQP-compatible position among the encodings we measured. On these images the reference protocol, a core frozen on one anchor followed by sequential per-image adaptation, generates noise. Joint training repairs it. A learned linear latent map then loses coherence past $M{=}6$, so a small non-linear map carries the latent instead. With eleven training images the resulting 10-qubit machine matches a DCGAN with six times as many parameters on both generation metrics within the seed spread. It leads the DCGAN on a flux-aligned variant, leads an RBM on the two MMD^2 columns, is level with the RBM on the histogram metric, and projects unseen images at correlation 0.901. With 100 images the DCGAN leads and the correlation reaches 0.977. All circuits are simulated exactly at $n = 10$ qubits over five seeds, so we claim no quantum advantage and treat the work as an early test of whether such a model could be worth deploying where training data are scarce.

1 Introduction

IQP circuits are commuting-gate circuits whose output distribution is the squared Walsh–Hadamard [30] transform of a phase function. Exact or multiplicatively approximate classical sampling from them would collapse the polynomial hierarchy [5], and the hardness is conjectured to hold on average under additive error [6]. The motivation has limits. Noise makes such circuits simulable in polynomial time [23], classical algorithms keep improving [18], and trainability itself pulls towards simulability [7]. The same structure keeps the expectation values of an MMD loss computable [25], as for other encodings built to stay simulable [17]. Such a model can therefore be trained classically and deployed on quantum hardware [24], following earlier Born machines trained with an MMD loss [16, 3, 9].

Whether real data sit where this helps is an empirical question. A recent proposal [15] approaches it with a *correlation–complexity map*. The map places a dataset at I_{QCLI}, the Jensen–Shannon divergence between its Walsh order spectrum (the Fourier power at each Hamming weight) and the i.i.d. binomial baseline, and at $I_{\mathrm{CCI}} \in [0, 1]$, the share of its correlation that the best pairwise tree misses [31, 26, 8]. IQP circuits should do well in the high-I_{QCLI}, high-I_{CCI} corner. The same paper gives a latent-adaptation protocol (Algorithm 1 of its v2), in which a phase core is trained on one anchor and a block of the first 50 phases in Hamming order is randomly initialised and adapted sample by sample. It was demonstrated at $n = 18$ qubits on 2D turbulence [13], one of several datasets the

Submitted to the 9th Workshop on Machine Learning and the Physical Sciences (ML4PS 2026). Do not distribute.

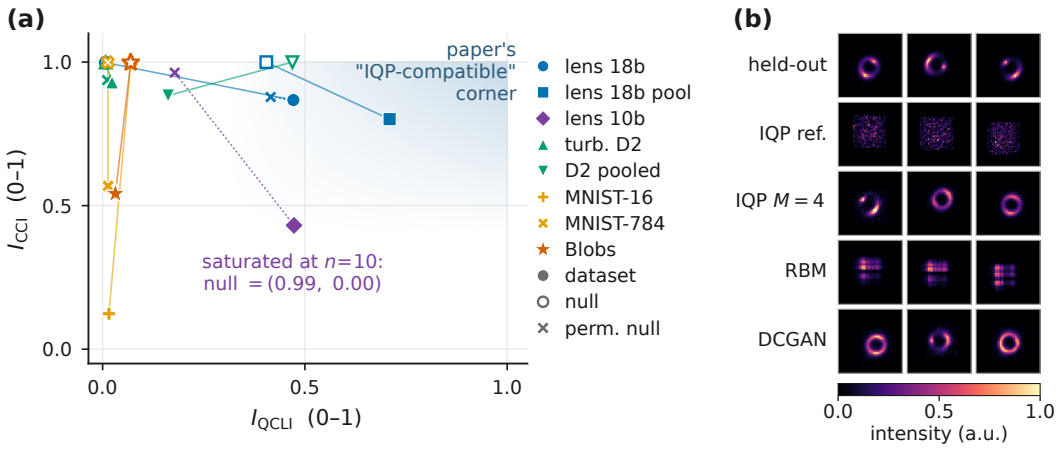


Figure 1: **(a)** Correlation–complexity map with datasets filled, i.i.d.-uniform nulls open and our permutation nulls as crosses. Lens 18b and 18b pool are the per-image and pooled 18-bit encodings, lens 10b is the crop pmf and turb. D2 is the reference turbulence data. **(b)** The first three unconditional samples per model at $N{=}100$, without selection.

map covers. A follow-up initialises the same fixed block from a physics-informed surrogate [14], whereas we learn the directions.

We apply both to simulated dark-matter strong-lensing images from DeepLense [1, 2]. Lens crops land at the most IQP-compatible map position we measured, and a permutation null shows that their I_{QCLI} is largely set by the intensity histogram, so that position is not by itself evidence of IQP-suitable structure. On static images the reference latent protocol produces noise. We trace the cause and replace its latent map by a non-linear one, then compare the model with an RBM and a DCGAN under two intensity conventions. App. C reports a replication on the turbulence data.

2 Setup

Four choices differ from the reference setup, each following from the data. The reference snapshots interpolate to unseen times, while the DeepLense `Model II` images we use [19] are static, so we generate unconditionally and project onto held-out images. Its 18-bit $(x, y, \mathrm{intensity})$ encoding suits a field of tuples, but the raw intensity axis of a lens crop carries barely one bit. We instead centroid-align each image and read its central 32×32 crop, which holds 99.4% of the light, as one probability mass function (pmf) over $1{,}024$ cells at $n = 10$ (App. A). The reference fixes the latent block, whereas we learn the map and train the N images jointly. It also estimates its MMD from samples, while a 10-qubit pmf is exact.

Every arm is a 10-qubit IQP Born machine over $1{,}024$ Walsh phases θ, one per qubit subset. The empty subset is a global phase, which leaves $1{,}023$ effective. The arms differ only in how a per-image latent enters. The reference protocol freezes the first 50 phases in Hamming order and adapts them sample by sample. On static crops this reconstructs the other training images poorly (Sec. 3.2), so we make the map trainable, first as a linear map of M directions,

$$\theta_i \;=\; \theta^0 + \Phi^\top z_i, \qquad \theta^0 \in \mathbb{R}^{1024}, \quad \Phi \in \mathbb{R}^{M\times 1024}, \quad z_i \in \mathbb{R}^M, \tag{1}$$

with each row of Φ normalised in L^2. On these crops the linear map degrades by $M{=}6$. We suspect the $\|z\|$ the images demand exceeds the radius over which the Walsh phases interfere coherently, but we have not isolated the cause. The main arm replaces $\Phi^\top z$ by a one-hidden-layer $\tanh$ map g of width $H{=}32$, with the z_i trained jointly with g under a small L^2 penalty. Its objective is the flat MMD (uniform Walsh weights, estimable from hardware shots). The three-bandwidth Gaussian kernel of [24], the *paper kernel* of the tables, is kept as a secondary arm. θ^0 is warm-started by $1{,}000$ exact-KL steps.

The trained pmf fixes only the relative intensities. We paste it into a 64×64 frame at a total intensity S and at an offset drawn from the offsets of the training crops. The reference convention gives every

frame the training-set mean S, so the generated flux has zero variance while real crops spread by about a quarter of their mean. We therefore draw S per image from a kernel density estimate of the N training totals, with no retraining, and App. A repeats everything under the fixed S. The baselines are a restricted Boltzmann machine (RBM) [28, 12, 29] and a deep convolutional generative adversarial network (DCGAN) [22]. Both are early-stopped on 200 validation images, while the IQP arms run a fixed $4{,}000$ steps, which can only favour the baselines. PDF–JS is the Jensen–Shannon divergence between intensity histograms. Conv2D-MMD^2 is an MMD in the feature space of a fixed random encoder [11, 4]. The arms set flux in different ways, so we add a post-hoc flux-aligned Conv2D-MMD^2 that rescales every frame to the held-out mean total. For unseen images we report the projection correlation, taking the best of eight latent restarts by the fit's own loss. Bins, scale and bandwidth are set on the held-out data alone, so no metric moves with the arm scored. Every arm generates 200 images against the same 200 held-out frames at $N \in \{11, 100\}$. Each number is a mean $\pm$ sample sd over five seeds, shown beside a real-vs-real floor between two disjoint real sets. The N training images are fixed across seeds, so the spread is initialisation and sampling variance alone. A lead in sd is the difference of two means in units of the baseline's seed sd, a descriptive quantity and not a significance test.

3 Results

3.1 Where lens crops sit on the correlation–complexity map

Table 1 recomputes the map (Fig. 1(a)). Under the encoding the reference paper uses for turbulence, lens images sit far above turbulence in I_{QCLI} and inside the corner the map calls IQP-compatible. A permutation null that keeps each image's intensity histogram comes within 14% of the lens value, and within a factor of two of the much smaller turbulence value. Under this encoding the index therefore mostly tracks the histogram. Both indices depend on K, the number of codes per image, so each row should be read against its own-K null. At the protocol's own $N_t = 5 \times 10^4$ samples per snapshot [15] the turbulence I_{QCLI} falls to 0.091, against a null of 0.098. The 10-bit crop pmf, which is the encoding we model, separates from its null in both coordinates. It sits at $I_{\mathrm{CCI}} = 0.43$, outside the high-I_{CCI} corner, so its high I_{QCLI} motivates the encoding but gives no reason to prefer an IQP over a pairwise model.

3.2 From the reference protocol to a non-linear latent map

Run sequentially on our own 50-phase block (App. C), Algorithm 1 produces noise (Fig. 1(b), second row). The circuit can represent the data, since it fits a *single* aligned crop to correlation 0.99997 ± 0.00001. The block is not the cause either, since training the same 50 phases jointly with the core across all N images (the *fixed basis, joint* row of Table 2) recovers most of the gap. The failure lies in freezing the core on one anchor and adapting each image in sequence. With the core frozen, the 50 fixed coordinates reconstruct the other training images at only 0.44 ± 0.04. We did not sweep the block size of 50. Equation (1) closes the rest of the gap but has a width limit of its own. On a 20-image sweep (Table 3) the held-out projection correlation of the learned linear map falls steeply from $M{=}2$ to $M{=}8$. The non-linear map avoids that collapse and stays slightly above the nearest-training-image reference. The non-linear map is flat to $M{=}4$ and degrades from $M{=}8$, so it does not buy a large useful width. At $M{=}2$, where we report it, its held-out projection correlation is 0.901 at $N{=}11$ and 0.977 at $N{=}100$. Projecting the nearest training image gives 0.892 and 0.970, and a single zero-initialised restart gives 0.838 and 0.887.

Table 1: Correlation–complexity map over five seeds. Each permutation null keeps the value histogram of the row above it, and ± 0.0000 marks a seed spread below 5×10^{-5}.

Dataset and encoding	K	I_{QCLI}	I_{CCI}
Lens, 18-bit $(x, y, \mathrm{intensity})$ tuples	4,096	0.4715 ± 0.0000	0.8677 ± 0.0000
permutation null	4,096	0.4154 ± 0.0000	0.8783 ± 0.0001
Turbulence D2, 18-bit tuples	4,096	0.0232 ± 0.0002	0.9285 ± 0.0008
permutation null	4,096	0.0114 ± 0.0000	0.9371 ± 0.0009
Lens, 10-bit crop pmf	exact pmf	0.4731 ± 0.0000	0.4313 ± 0.0000
permutation null	exact pmf	0.1782 ± 0.0000	0.9634 ± 0.0004

Table 2: Lens crops under the sampled intensity over five seeds, with PDF–JS in 10^{-3} and both MMD^2 columns in 10^{-2} (lower is better). **Bold** marks a best mean whose gap to every other arm exceeds the sum of the two seed sds, so a column without bold has no win the seed spread supports. Params are totals at N=11 / N=100. App. C defines † and the parameter convention. The last two rows give the lead of the non-linear flat-MMD arm at N=11 in the baseline's seed sd (Sec. 2).

		$N = 11$			$N = 100$		
Model	Params	PDF–JS	MMD^2	MMD^2_{al}	PDF–JS	MMD^2	MMD^2_{al}
IQP, reference protocol (Alg. 1)†	1,524 / 5,974	6.5 ± 0.8	14.8 ± 1.6	22.5 ± 2.2	7.7 ± 0.4	14.7 ± 0.3	26.6 ± 0.7
IQP, fixed basis, joint	1,524 / 5,974	1.8 ± 0.4	8.7 ± 1.4	12.5 ± 1.3	1.1 ± 0.2	5.4 ± 0.4	8.5 ± 1.5
IQP, linear map M=4 (flat MMD)	5,164 / 5,520	2.8 ± 0.1	5.6 ± 1.9	2.7 ± 0.2	2.6 ± 0.2	2.8 ± 0.5	2.6 ± 0.4
IQP, linear map M=4 (paper kernel)	5,164 / 5,520	1.0 ± 0.05	6.3 ± 1.5	7.4 ± 0.6	1.5 ± 0.2	5.9 ± 0.4	10.0 ± 0.8
IQP, non-linear map M=2 (flat MMD)	33,910 / 34,088	1.2 ± 0.07	5.2 ± 1.8	$\mathbf{2.2} \pm 0.2$	1.7 ± 0.2	2.4 ± 0.5	1.8 ± 0.2
IQP, non-linear map M=2 (paper kernel)	33,910 / 34,088	0.75 ± 0.07	5.4 ± 1.5	5.1 ± 0.3	0.99 ± 0.04	4.2 ± 0.4	6.6 ± 0.6
RBM [29]	11,510 / 15,960	1.9 ± 1.0	7.9 ± 1.9	11.1 ± 3.9	1.1 ± 0.2	5.1 ± 0.8	8.1 ± 2.2
DCGAN [22]	199,837	1.1 ± 0.6	5.1 ± 1.5	3.9 ± 1.0	$\mathbf{0.55} \pm 0.12$	2.2 ± 0.3	2.1 ± 0.4
Real-vs-real floor	n/a	*0.04* ± 0.02	*0.5* ± 0.3	*0.4* ± 0.2	*0.04* ± 0.02	*0.5* ± 0.3	*0.4* ± 0.2
Lead over DCGAN, in DCGAN sd		-0.13	-0.04	$+1.57$			
Lead over RBM, in RBM sd		$+0.75$	$+1.40$	$+2.27$			

3.3 Comparison with classical baselines

The IQP arms are competitive in the eleven-image regime of Table 2. There the non-linear arm matches the DCGAN on PDF–JS and raw Conv2D-MMD^2 within seed noise and leads on the flux-aligned column. It leads the RBM on the two MMD^2 columns and is within one sd on PDF–JS (last two rows of Table 2), with a sixth of the DCGAN's parameters and three times the RBM's. Under the paper kernel the same map has the lowest mean PDF–JS, 0.75×10^{-3} against 1.1 ± 0.6 for the DCGAN, still within one sd, and a worse flux-aligned column. At neither size does an arm lead the DCGAN on all three columns by more than a seed sd. At 100 images the DCGAN leads on PDF–JS and raw Conv2D-MMD^2, and the flux-aligned 1.8 ± 0.2 sits within one sd of its 2.1 ± 0.4.

The flux-aligned column does not depend on the intensity convention, and the largest paired difference over the 325 runs carrying both is 6×10^{-9}. Invariance is not independence, and the two MMD^2 columns overlap, since the aligned one measures shape and the raw one measures shape plus flux. The sampling rule also treats the arms unequally. At N=11 it improves the raw Conv2D-MMD^2 of every IQP arm by 22 to 39% and makes the RBM's 20% worse, and under the fixed intensity the RBM leads the non-linear arm on that column (6.5 ± 0.8 against 8.5 ± 0.2, App. A). The raw-column ranking against the RBM is therefore a property of the sampled decode and not of the model. Sampling reproduces the spread of the training totals but keeps their mean, and the eleven training frames are 9% dimmer than the held-out set. That offset is why the raw column of the learned-map IQP arms at N=11 lands level with the DCGAN and not near the floor. App. B reports data scaling over $N \in \{10, \ldots, 100\}$ and per-class retraining.

4 Discussion and limitations

The competitive standing of the 10-qubit model at eleven images does not carry over to 100 images. A deployed margin would also be bounded by connectivity and noise [21]. The results concern sample quality and projection fidelity alone and do not show that a quantum device would beat a classical one on this task. Generated frames sit farther in L_2 from their nearest training image than real held-out frames do (4.21 ± 0.10 against 2.90 at N=11), so the arm does not memorise. Two steps cannot be estimated from shots as run. The exact-KL warm start can be replaced within one seed sd (App. C), but the projection optimises z against the exact pmf. At $n = 10$ we parameterise all 2^n Walsh phases, and scaling up would need them sparsified [15], which we have not tested. Only the model fit sees N images. Bandwidths, floors and the baselines' early stopping use hundreds to thousands of real frames, so the study limits the training set and not the pipeline. Alignment and the crop leave translation and the outer 0.6% of the light outside the task, and the floor shares this restriction. Five seeds are few. The DCGAN alone is not bit-reproducible and adds up to 57% of spread at one seed. The sampled intensity moves the raw column unequally across arms. Everything

rests on one simulated dataset, the metrics do not test substructure, and no destroyed image is scored as an upper reference.

References

[1] Stephon Alexander, Sergei Gleyzer, Evan McDonough, Michael W. Toomey, and Emanuele Usai. Deep learning the morphology of dark matter substructure. *The Astrophysical Journal*, 893(1):15, 2020.

[2] Stephon Alexander, Sergei Gleyzer, Hanna Parul, Pranath Reddy, Michael W. Toomey, Emanuele Usai, and Ryker Von Klar. Decoding dark matter substructure without supervision, 2020. arXiv:2008.12731.

[3] Marcello Benedetti, Delfina Garcia-Pintos, Oscar Perdomo, Vicente Leyton-Ortega, Yunseong Nam, and Alejandro Perdomo-Ortiz. A generative modeling approach for benchmarking and training shallow quantum circuits. *npj Quantum Information*, 5(1):45, 2019.

[4] Mikołaj Bińkowski, Danica J. Sutherland, Michael Arbel, and Arthur Gretton. Demystifying MMD GANs. In *International Conference on Learning Representations (ICLR)*, 2018.

[5] Michael J. Bremner, Richard Jozsa, and Dan J. Shepherd. Classical simulation of commuting quantum computations implies collapse of the polynomial hierarchy. *Proceedings of the Royal Society A*, 467(2126):459–472, 2011.

[6] Michael J. Bremner, Ashley Montanaro, and Dan J. Shepherd. Average-case complexity versus approximate simulation of commuting quantum computations. *Physical Review Letters*, 117:080501, 2016.

[7] M. Cerezo, Martin Larocca, Diego García-Martín, N. L. Diaz, Paolo Braccia, Enrico Fontana, Manuel S. Rudolph, Pablo Bermejo, Aroosa Ijaz, Supanut Thanasilp, Eric R. Anschuetz, and Zoë Holmes. Does provable absence of barren plateaus imply classical simulability? *Nature Communications*, 16(1):7907, 2025.

[8] C. K. Chow and C. N. Liu. Approximating discrete probability distributions with dependence trees. *IEEE Transactions on Information Theory*, 14(3):462–467, 1968.

[9] Brian Coyle, Daniel Mills, Vincent Danos, and Elham Kashefi. The Born supremacy: quantum advantage and training of an Ising Born machine. *npj Quantum Information*, 6(1):60, 2020.

[10] Kaitlin Gili, Marta Mauri, and Alejandro Perdomo-Ortiz. Generalization metrics for practical quantum advantage in generative models. *Physical Review Applied*, 21(4):044032, 2024.

[11] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773, 2012.

[12] Geoffrey E. Hinton. Training products of experts by minimizing contrastive divergence. *Neural Computation*, 14(8):1771–1800, 2002.

[13] Ali Rahimi Khojasteh, Sylvain Laizet, Dominique Heitz, and Yin Yang. Lagrangian and Eulerian dataset of the wake downstream of a smooth cylinder at a Reynolds number equal to 3900. *Data in Brief*, 40:107725, 2022.

[14] Chen-Yu Liu, Leonardo Placidi, Marco Ballarin, and Enrico Rinaldi. Generative IQP circuit learning with physics-informed latent initialization, 2026. arXiv:2607.28866.

[15] Chen-Yu Liu, Leonardo Placidi, Eric Brunner, and Enrico Rinaldi. Toward generative quantum utility via correlation–complexity map, 2026. arXiv:2603.06440v2 (v2, 5 May 2026); all section, equation, proposition and algorithm numbers refer to v2.

[16] Jin-Guo Liu and Lei Wang. Differentiable learning of quantum circuit Born machines. *Physical Review A*, 98(6):062324, 2018.

[17] Daniele Lizzio Bosco, Gabriel Matos, Chen-Yu Liu, Frederic Rapp, Fabian Finger, Enrico Rinaldi, and Konstantinos Meichanetzidis. Efficiently simulable Pauli correlation encoding, 2026. arXiv:2607.20409.

[18] Dmitri Maslov, Sergey Bravyi, Felix Tripier, Andrii Maksymov, and Joe Latone. Fast classical simulation of Harvard/QuEra IQP circuits, 2024. arXiv:2402.03211.

[19] ML4SCI Collaboration. DeepLense: Deep learning for strong gravitational lensing. `https://github.com/ML4SCI/DeepLense`, 2025.

[20] Mustafa Mustafa, Deborah Bard, Wahid Bhimji, Zarija Lukić, Rami Al-Rfou, and Jan M. Kratochvil. CosmoGAN: creating high-fidelity weak lensing convergence maps using generative adversarial networks. *Computational Astrophysics and Cosmology*, 6(1):1, 2019.

[21] Leonardo Placidi, Enrico Rinaldi, Keisuke Fujii, and Chen-Yu Liu. The impact of qubit connectivity on quantum advantage in noisy IQP circuits, 2026. arXiv:2604.12635. Accepted at IEEE QCE 2026.

[22] Alec Radford, Luke Metz, and Soumith Chintala. Unsupervised representation learning with deep convolutional generative adversarial networks, 2015. arXiv:1511.06434.

[23] Joel Rajakumar, James D. Watson, and Yi-Kai Liu. Polynomial-time classical simulation of noisy IQP circuits with constant depth. In *Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1037–1056, 2025.

[24] Erik Recio-Armengol, Shahnawaz Ahmed, and Joseph Bowles. Train on classical, deploy on quantum: scaling generative quantum machine learning to a thousand qubits, 2025. arXiv:2503.02934.

[25] Erik Recio-Armengol and Joseph Bowles. IQPopt: Fast optimization of instantaneous quantum polynomial circuits in JAX, 2025. arXiv:2501.04776.

[26] Elad Schneidman, Susanne Still, Michael J. Berry, and William Bialek. Network information and connected correlations. *Physical Review Letters*, 91(23):238701, 2003.

[27] Bernard W. Silverman. *Density Estimation for Statistics and Data Analysis*. Chapman and Hall, London, 1986.

[28] Paul Smolensky. Information processing in dynamical systems: foundations of harmony theory. In David E. Rumelhart and James L. McClelland, editors, *Parallel Distributed Processing: Explorations in the Microstructure of Cognition, Volume 1: Foundations*, pages 194–281. MIT Press, Cambridge, MA, 1986.

[29] Tijmen Tieleman. Training restricted Boltzmann machines using approximations to the likelihood gradient. In *Proceedings of the 25th International Conference on Machine Learning (ICML)*, pages 1064–1071, 2008.

[30] Joseph L. Walsh. A closed set of normal orthogonal functions. *American Journal of Mathematics*, 45(1):5–24, 1923.

[31] Satosi Watanabe. Information theoretical analysis of multivariate correlation. *IBM Journal of Research and Development*, 4(1):66–82, 1960.

A Pipeline and the fixed-S convention

We reran Table 2 with the single training-mean intensity under which every pmf-based arm was published. The same runs rebuild the published frames exactly, so the two decodes are paired. At $N{=}11$ sampling improves the raw Conv2D-MMD^2 of every IQP arm by 0.033 to 0.041 (paired sd 0.012 to 0.020) and makes the RBM worse by 0.013 ± 0.017, which explains the RBM lead under the fixed intensity reported in the main text. At $N{=}100$ sampling improves every arm. The PDF–JS effect is at most 7×10^{-4} and mixed in sign, and the flux-aligned effect is below 10^{-8}. The full table is in the released seed tables.

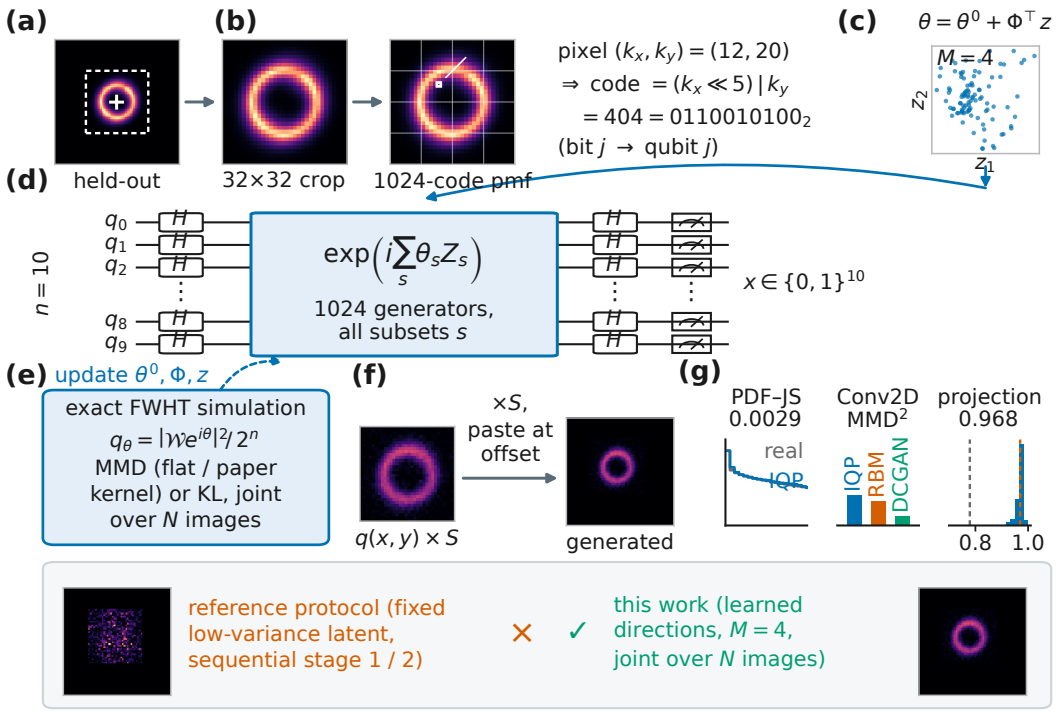


Figure 2: Pipeline. **(a)** A held-out image and the crop window. **(b)** The aligned crop as a pmf over 1,024 cells. Pixel $(12, 20)$ carries code 404, and bit j addresses qubit j. **(c)** Learned latents at $M = 4$. **(d)** The circuit. **(e)** Exact simulation, drawn for Eq. (1). The main arm holds $\theta = \theta^0 + g(z)$. **(f)** Decoding at S. **(g)** Metrics for the linear $M{=}4$ arm.

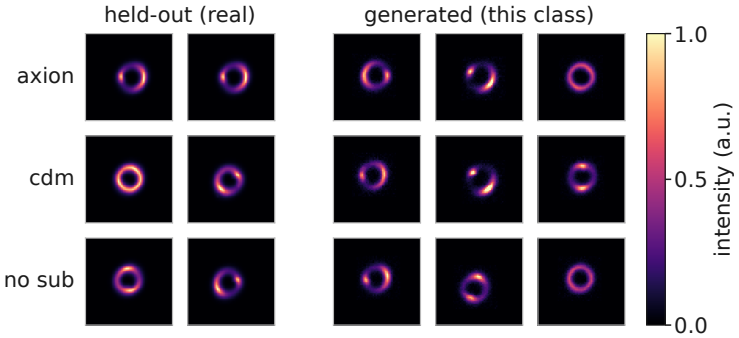


Figure 3: One row per substructure class at $N{=}100$, on a shared 0 to 1 intensity scale. Left, two held-out frames of that class. Right, three unconditional samples from the non-linear $M{=}2$ arm retrained on that class alone, taken as the first two and first three in index order without selection, seed 0.

B Latent sweeps and per-class results

Hidden width matters less than latent width, since every width from $H{=}16$ up lies inside a seed sd. The latent-width sweep is flat over $M \leq 4$, degrades at $M{=}8$ and collapses at $M{=}16$. Both full sweeps are in the released seed tables. In Table 4 the raw Conv2D-MMD^2 of the non-linear arm and the RBM falls with N, and the DCGAN starts ahead and ends level. On the flux-aligned column the non-linear arm is lowest at every N, but by more than a seed sd only at $N{=}10$ and $N{=}70$. Retraining inside one class (Table 5, Fig. 3) shows that each class can be generated on its own. It does not show that one model separates the classes. At $N{=}11$ the mean $\|z\|$ is 0.101 at $M{=}2$ and 0.120 at $M{=}8$, while the affine residual of g falls from 0.70 to 0.17, so g is closest to linear where it fails.

Table 3: Projection correlation, 20-image sweep. The two maps use separate drivers, so a small gap between them means nothing.

Latent map	M	Correlation
linear	2	0.906 ± 0.002
linear	6	0.738 ± 0.067
linear	8	0.218 ± 0.121
non-linear	all	0.9368 to 0.9519
nearest train. image	n/a	0.9339

Table 4: Data scaling over five seeds, sampled intensity, raw units. Blank cells are not reported. The RBM's flux-aligned value spans 0.077 to 0.127.

	MMD^2			MMD$^2_{\text{a1}}$
N	non-linear	RBM	DCGAN	non-linear
10	0.062 ± 0.016	0.091 ± 0.017	0.046 ± 0.017	0.026
30				0.023
50				0.020
70				0.016
90				0.019
100	0.024 ± 0.005	0.051 ± 0.008		0.018

Table 5: Per class, each arm retrained on that class alone and scored against that class's own 200 held-out frames, under the sampled intensity over five seeds, in the units of Table 2. Values compare within a class only. The real-vs-real floors are $0.03/0.36/0.31$ for axion, $0.03/0.30/0.32$ for cdm and $0.04/0.45/0.45$ for no sub, in the same units and order.

		non-linear $M{=}2$			linear $M{=}4$		
Class	N	PDF–JS	MMD^2	MMD$^2_{\text{a1}}$	PDF–JS	MMD^2	MMD$^2_{\text{a1}}$
axion	11	1.1 ± 0.10	4.6 ± 0.6	2.9 ± 0.4	3.4 ± 0.20	4.7 ± 0.7	2.8 ± 0.5
axion	100	1.8 ± 0.17	2.9 ± 0.2	1.7 ± 0.2	2.6 ± 0.11	2.7 ± 0.2	2.1 ± 0.2
cdm	11	0.94 ± 0.05	4.4 ± 0.2	2.7 ± 0.3	2.4 ± 0.09	4.3 ± 0.3	2.6 ± 0.5
cdm	100	1.7 ± 0.06	2.1 ± 0.3	1.5 ± 0.3	2.7 ± 0.25	2.3 ± 0.4	1.8 ± 0.3
no sub	11	0.98 ± 0.10	3.1 ± 0.9	2.4 ± 0.6	2.6 ± 0.22	3.5 ± 0.8	2.6 ± 0.1
no sub	100	1.7 ± 0.10	2.3 ± 0.4	2.1 ± 0.3	2.7 ± 0.31	2.7 ± 0.4	2.9 ± 0.3

C Turbulence replication, training and evaluation details

On that paper's own dataset (2D cylinder-wake turbulence D2 [13], $n = 18$ qubits, 11 anchor snapshots, five seeds) we do not reproduce the result under our pre-registered convention. The RBM (PDF–JS 0.217 ± 0.007) and a 1.2M-parameter DCGAN (0.045 ± 0.062) both score better than our 31k-parameter IQP (0.272 ± 0.004) on both metrics. Under last-write-wins aggregation, which that paper [15] never specifies, our PDF–JS moves towards the published values without reaching them (0.0324 and 0.0473 against 0.0238 and 0.0221), and the RBM still leads. On these data Conv2D-MMD^2 does not rank a destroyed image below a trained one. Pixel-shuffled ground truth scores 0.7045 ± 0.0095, better than every model in that paper's Table 2. That paper itself notes that its baselines are not exhaustive and that this metric does not replace spectral diagnostics [15].

The † arm runs Algorithm 1 of v2 [15] on our own latent block. That block holds the 50 subsets of Hamming weight at most 2 whose Walsh coefficients vary most over a disjoint pool of 2,000 images, where the reference block is the first 50 in Hamming order. *Fixed basis, joint* trains that block jointly with the core across all N images. Params are totals at that N, counting shared parameters (θ^0 included) plus N per-image latents. The fixed-basis arms hold $974 + 50N$, the learned-direction arms $5{,}120 + MN$ and the RBM $10{,}960 + 50N$. The DCGAN entry counts its generator alone, and its discriminator adds $53{,}497$. The RBM has 10 visible and $1{,}000$ hidden units on the 10-bit code. Each image owns its visible biases and the first 40 hidden biases, and the RBM ran 160k or 80k steps. The DCGAN builds the crop from a 128-dimensional prior through four transposed convolutions, ran 5k to 30k steps and learns its own intensity.

Table 6: Wasserstein-1 distance in ring radius r and ellipticity e_1, e_2. Blank cells are not reported.

	$N = 11$			$N = 100$		
Arm	r	e_1	e_2	r	e_1	e_2
main arm	0.82	0.33	0.35	1.01	0.31	0.14
RBM	3.23	0.46	0.24			
DCGAN	1.67	0.62	0.40	0.40	0.28	0.15
Real-vs-real	*0.10*	*0.17*	*0.10*			

The paper kernel is the three-bandwidth Gaussian MMD of [24] at $\sigma = \{0.989, 1.729, 2.236\}$. The protocol we replicate publishes no bandwidths for this encoding, so we set σ_1 to make the average Pauli weight 2, σ_3 to the square root of the median heuristic and σ_2 to their root mean square. The Conv2D-MMD^2 bandwidth is 0.005547 raw and 0.002558 flux-aligned. It is fixed once on the held-out features and shared by every table except the per-class rows. The 20 floor draws come from the $8{,}710$ validation images the held-out split leaves. The bandwidth of the intensity estimate, by Silverman's rule of thumb [27], is 14.64 at $N{=}11$ and 10.36 at $N{=}100$.

Applied to the IQP arms, the baselines' validation rule selects 2k to 16k steps. Every Table 2 cell it changes moves toward better IQP values and no comparison flips, so our pre-registered $4{,}000$ steps is the conservative choice. At $20{,}000$ fixed steps both baselines are much worse, so their early stopping matters and we keep it. None of three ablations moves a generation metric by more than a seed sd. A flat-MMD warm start of θ^0, which makes every stage shot-estimable, leaves the flux-aligned column at 0.0231 against 0.0225 at $N{=}11$. Picking the projection restart by correlation instead of fit loss moves the correlation by 0.0003, and two restarts match eight. At $M{=}2$ projecting the mean crop gives 0.765 and 0.779 at the two training-set sizes. Our Conv2D-MMD^2 is the kernel inception distance [4] with an untrained encoder. We do not use the purpose-built generalisation metrics of [10], and classical generation of cosmological images is established [20]. In ring morphology (Table 6) the main arm is closest in radius at $N{=}11$ and the DCGAN is closest at $N{=}100$. Retraining the same DCGAN configuration at the same seed moves its raw Conv2D-MMD^2 by 5 to 57% and its PDF–JS by 2 to 74%, and in this sense it is the one arm that is not bit-reproducible.

D Qubit count, the reference paper's Fig. 4, and the departures in full

The qubit count has to match the effective resolution of the image, a choice the reference paper does not face with a field of tuples at $n = 18$. At $n = 8$ the 16×16 pmf averages the ring away and the fixed bilinear upsampling cannot restore it, so the flux-aligned error is more than twice the $n = 10$ value. At $n = 12$ about three quarters of the $4{,}096$ cells carry background rather than lens light. The arm learns that background and becomes unstable across seeds, and its $N{=}100$ PDF–JS reaches 92 ± 71. We therefore work at $n = 10$.

Table 7: Qubit ladder at $N{=}11$ over five seeds, in the units of Table 2. The flux-aligned column does not depend on the intensity convention (App. A) and compares across rows. The other two columns use the fixed intensity, except in the $n = 8$ row (marked *). The latent map differs by row, so the table is a ladder of encodings and does not vary n alone.

n	Encoding	Latent map	Params	PDF–JS	MMD^2	MMD$^2_{\text{al}}$
8	16×16 pmf, bilinear upsampling	MLP $H{=}8$, $M{=}2$	2,350	$1.7 \pm 0.05^*$	$8.0 \pm 1.7^*$	5.1 ± 0.3
10	32×32 aligned crop pmf	MLP $H{=}32$, $M{=}2$	33,910	1.0 ± 0.1	8.5 ± 0.2	$\mathbf{2.3} \pm 0.2$
12	whole 64×64 frame as one pmf	linear, $M{=}4$	20,524	43 ± 30	23.3 ± 8.8	18.4 ± 11.3

We re-ran that paper's Fig. 4 over five seeds of SPSA (simultaneous perturbation stochastic approximation, which that paper uses to spread its targets over I_{QCLI}). Its positive coupling at low I_{QCLI} ($I_{\text{QCLI}} \leq 0.3$) reproduces at every size and in every seed, with Spearman $+0.92 \pm 0.10$, $+0.91 \pm 0.04$ and $+0.96 \pm 0.01$ at $n = 8$, 12, 16. Above that range the sign of the global coefficient changes from seed to seed.

Table 8: Each departure from the protocol we replicate, with the property of the data that drove it. None of them came from tuning.

	Reference protocol	This paper	Property of the data
Encoding	18-bit $(x, y, \text{intensity})$ tuples, many per snapshot	one 10-bit pmf over $1{,}024$ cells of an aligned 32×32 crop	the raw intensity axis carries barely one bit and 99.4% of the light is inside the crop
Latent	a fixed block of 50 phases, the first in Hamming order	a learned map, linear $\Phi^\top z$ or a $\tanh$ $g(z)$	on static images fixed directions cross-fade pixels instead of moving ring structure
Training	core fitted on one anchor, then each sample adapted in sequence	core and all N latents trained jointly, $4{,}000$ Adam steps	with at most 100 static frames there is no stream to adapt along
Objective	multi-bandwidth Gaussian MMD estimated from samples	flat MMD as the main arm, that kernel kept as a secondary one	at $n = 10$ the pmf is exact, so sampling noise would be a choice
Flux	every decoded frame carries the training-set mean total	S drawn per image from a density estimate of the N totals	real crops spread by about a quarter of their mean, which one S cannot reproduce
Evaluation	held-out times, that paper's own metrics	200 held-out frames fixed across arms and seeds, bins and bandwidth set on them alone	the images are static, so time cannot be held out
Baselines	compared at a fixed budget	RBM and DCGAN early-stopped on 200 validation frames, IQP arms not	the baselines converge at very different step counts, so one budget would favour us